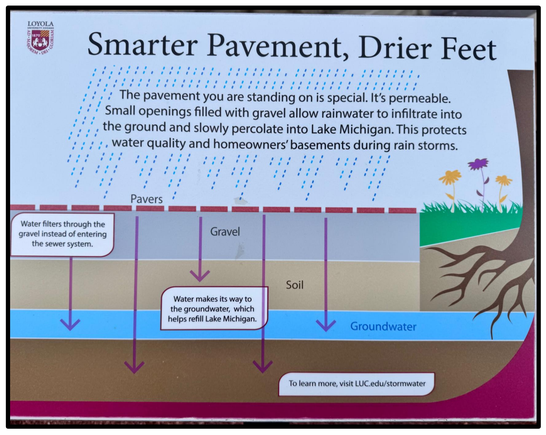
Heels vs. Permeable Pavement

Layla Tasker, Jayla Davis, Mekhi Morgan, and Noel Serrano, Department of Engineering
Sponsor and Faculty Advisor: Dr. Gail Baura

Problem Definition & Constraints

Heels, worn by students and teachers, are susceptible to getting stuck and/or damaged when caught in between the permeable pavers on campus. We were tasked to make something to solve this issue.



Realistic Constraints:

1. Budget ($100)

2. Materials/Design (must accommodate different heel sizes

Materials

Material	Price
ABS Plastic	$0.32 per device ($0.65 for both)

Experimental Measurements & Analysis

Triangle Design

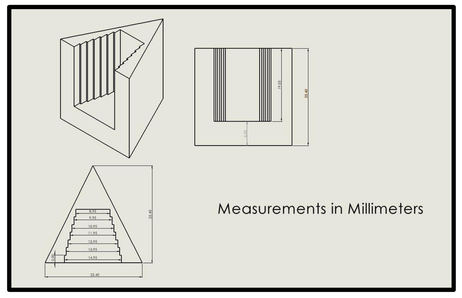
Length: 25 mm
Width: 25 mm
Height: 1 in (25.4mm)
Notch Depth: ¾ in (19.05mm)
Side length per notch: 2.5mm.
Width per notch (biggest to smallest):
-14.95mm - 8.95mm

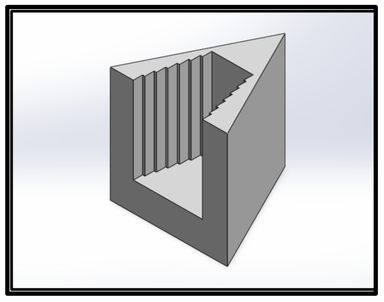
Horseshoe Design

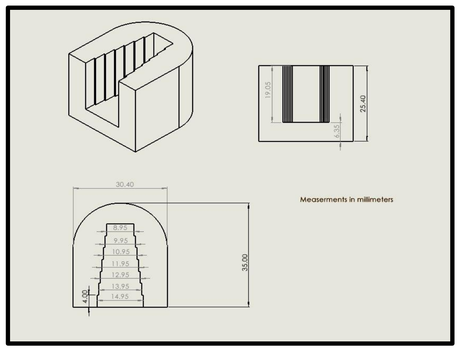
Length: 25 mm
Width: 25 mm
Height: 1 in (25.4mm)
Notch Depth: ¾ in (19.05mm) *
Side length per notch: 2.5mm.
Width per notch (biggest to smallest) [1]:
-14.95mm - 8.95mm

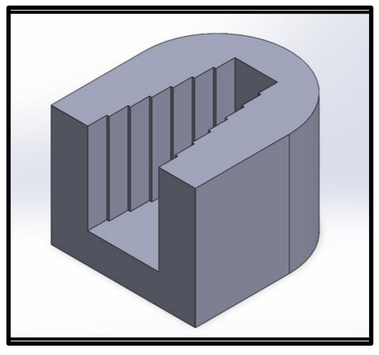
This is the same depth for all notches from the top of the apparatus to the bottom of the apparatus

Feasibility of Proposed Solutions









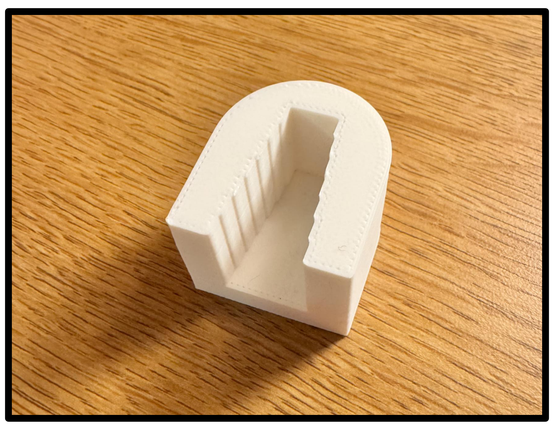
Usage of designs:

1.You will slide the bottom of the heel into the desired heel size notch

2.Wedge it into the corresponding notch securely

The sides of the apparatus would prevent any torque that may occur when turning while walking, along with creating static friction with the heel itself to prevent the heel from slipping out of the apparatus in general.

Model Creation



Creating model:

Took the solidwoks designs and printed the horseshoe design using a 3D printer and ABS Plastic.

(creation took a while due to the hours of 3D printing)

Recommended Solution



Survey Questions (19 people responded):

1. Which design do you prefer?
2. Could you see yourself wearing it?
3. How well do you think this could work?
4. What scenario could you see this not working?
5. If there is anything you would change about the design? If so, what would it be?

Testing:
Our sponsor tried on the shoe with the horseshoe design and tested the product while walking

Better Design: Horseshoe

- like the original design of the heel
- more aesthetically pleasing than the triangle design
- survey results sided with the design more

Conclusion

After designing and testing, we decided that the horseshoe design would be the best solution to prevent heels from being wedged in permeable pavement. With this design the user will be able to protect their shoes without taking from the aesthetic of them.

References

[1] Belgard Architectural Products Group. (2018). High heel shoes & belgard® permeable pavers. https://www.belgardcommercial.com/wpcontent/uploads/2023/10/Belgard-Technical-Note-High-Heel-Shoes-and-Belgard-Permeable-Pavers.pdf